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Multi-line LiDAR 3D construction environment modeling and BIM consistency update method for digital twins

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Construction digital twins aim to maintain an up-to-date, BIM-consistent as-built state from streaming multi-line LiDAR to enable progress verification, dimensional compliance control, and proactive safety management on construction sites. However, converting sequential LiDAR observations into a temporally stable, element-centric BIM state is difficult in realistic AEC environments. Existing approaches mainly fall into geometry-driven Scan-to-BIM registration frameworks and learning-based point cloud perception models, with some system-level digital-twin frameworks built on BIM-centric reasoning. Purely geometric alignment is brittle under occlusion, clutter, and repetitive structures, and registration uncertainty can propagate to downstream progress and deviation conclusions. Meanwhile, generic perception networks are often trained on static frames, lack temporal stabilization, and do not explicitly optimize BIM-consistent element association or geometry-aware alignment cues. In addition, many digital-twin systems expose limited, loosely coupled state representations that hinder causal, reliable site updates. To address these gaps, we propose LiBiDT (LiDAR–BIM Digital Twin), a causal LiDAR–BIM digital-twin model—where "causal" refers to computational online updating in which only past observations enter the current state—that integrates spatio-temporal mapping, geometric–semantic perception, and scan-to-BIM association for consistent state evolution. LiBiDT consists of Spatio-Temporal Global Registration (STGR) with LiDAR–IMU–GNSS priors, Multi-task Perception (MTP) using a PointNet++ backbone with semantic/instance/normal heads, and Scan-to-BIM Association and State Updating (S2B) via normal-guided point-to-plane refinement, bipartite matching with fused overlap/deviation/semantic evidence, and exponential moving average (EMA) stabilization. The Decision and Risk Analytics (DRA) layer turns the BIM-aligned state into auditable construction-management indicators—weighted completion, deviation summaries, and three forward-evaluable safety/compliance hazard flags. Experiments on three public benchmarks (Hilti SLAM, CV4AEC 2024 Scan-to-BIM, and S3DIS Area-5) cover trajectory, perception, and association evaluation. On CV4AEC 2024 Scan-to-BIM, LiBiDT reaches association F1 = 0.837 and Chamfer Distance (CD) / surface Root Mean Square Error (RMSE) of 3.2/3.5 cm; relative to the perception-replacement baseline (PointNet++ + S2B, F1 = 0.758) the gain is large, while relative to the strongest competing baseline—CoFiNet (Reg) + S2B at F1 = 0.822—the absolute association gain narrows to 0.015 F1 points, with the bigger gains coming from temporal identity consistency (IDF1 0.854 → 0.861) and ID-switch reduction. A held-out tile-stream case study visualises BIM-element state evolution, alignment refinement, EMA-stabilised evidence streams, and hazard-flag firing as the scan progressively covers the floor; a strict temporal validation under active construction will require time-stamped multi-acquisition data and is left to future work. Ablation results confirm that evidence stabilization and multi-cue matching contribute complementary gains in accuracy and temporal consistency, yielding a favorable efficiency–accuracy trade-off.

Keywords Construction digital twin, Multi-line LiDAR, Scan-to-BIM association, Point cloud perception, BIM consistency update

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Building and infrastructure projects increasingly rely on fine-grained, near-real-time site awareness to support progress verification, quantity takeoff and payment, dimensional compliance control, and proactive safety management. Building Information Modeling (BIM) provides a design-time, element-centric representation, but the as-built scene evolves continuously and often deviates from the planned model. Construction digital twins aim to close this loop by fusing sensing data with BIM to maintain an up-to-date, operationally interpretable state of the site[1–3]. Among available sensing modalities, multi-line LiDAR (Light Detection and Ranging) offers dense 3D geometry with robust depth measurements, enabling frequent capture of complex construction scenes; however, turning streaming LiDAR into a temporally consistent, BIM-aligned, element-level state remains challenging due to clutter and occlusion, repetitive structures, coordinate-frame drift, heterogeneous supervision for perception, and the need for causal, on-site operation.

A first family of approaches is geometry-driven Scan-to-BIM alignment and model-based comparison, typically centered on coarse-to-fine registration and handcrafted geometric cues. Representative Scan-to-BIM frameworks exploit planar primitives or congruent sets to initialize the alignment and then refine it[4]. While effective in controlled settings, these methods are brittle under partial observations, non-model objects, and symmetries that induce ambiguous alignments; moreover, uncertainty in geometric matching can propagate to downstream "progress/quality" conclusions when the digital twin is built from imperfect registrations[4]. More broadly, systematic reviews note that fully automated, robust Scan-to-BIM remains difficult in realistic Architecture, Engineering, and Construction (AEC) scenarios, especially when the environment is incomplete, cluttered, or only partially scanned[5].

A second family focuses on learning-based point cloud understanding. Deep networks for 3D segmentation and feature learning have significantly advanced geometric–semantic perception[6–8]. Construction-oriented studies further demonstrate that learned scene understanding can support Scan-to-BIM automation and downstream reconstruction[9]. Nonetheless, these models are commonly trained on static frames and do not, by themselves, enforce temporally stable predictions across a LiDAR stream, nor do they guarantee BIM-consistent element associations and geometry-aware alignment cues needed by Scan-vs-BIM workflows. In addition, instance-level consistency and robust clustering remain non-trivial in cluttered scenes even with discriminative embedding objectives[10].

A third family concerns digital-twin systems and construction management analytics (progress, quality, and safety) built on BIM-centric reasoning. Prior work highlights the promise of digital twins for cyber-physical integration but also emphasizes interoperability, real-time updating, and reliable data-to-model synchronization as persistent gaps[2,3]. On the safety side, BIM-based hazard identification and prevention can encode design-time priors and rules, yet its operational effectiveness hinges on accurate as-built localization of hazards and protections, which is difficult without a robust, continuously updated scan-to-BIM state[11]. Consequently, construction-management indicators often remain loosely coupled to the sensing and alignment stack, limiting reliability when scene observations are noisy, intermittent, or partially missing.

In summary, existing models provide valuable components, but they do not fully deliver a causal, BIM-consistent digital-twin update that tightly couples spatio-temporal mapping, geometric–semantic perception, element-level association, and construction-management indicators from LiDAR streams. The main problems with these methods are as follows:

- **Fragmented sensing-to-BIM state updating**: Many approaches stop at per-scan registration or per-frame segmentation, without an explicit element-level digital-twin state and a causal update mechanism that is robust to occlusion and intermittent observations[2,3].
- **Brittle scan-to-BIM association under real site conditions**: Purely geometric alignment and matching can be ambiguous in symmetric, cluttered, and partially observed scenes, and uncertainty in scan-vs-BIM geometry can distort downstream conclusions[4].
- **Limited integrated, geometry-aware state for site management**: Many systems do not expose a unified, BIM-aligned geometric–semantic state that can be updated online and used consistently across progress, dimensional-compliance, and deviation monitoring[11].

To address these limitations, we propose ***LiBiDT*** (*LiDAR–BIM Digital Twin*), a four-stage digital-twin model that performs *Spatio-Temporal Global Registration (STGR)*, *Multi-task Perception (MTP)*, *Scan-to-BIM Association and State Updating (S2B)*, and *Decision and Risk Analytics (DRA)* in a strict causal cascade. The key idea is to treat globally registered LiDAR as a scene-level backbone, attach geometric–semantic predictions to it, and update BIM element states through robust association and temporally filtered evidence, producing an auditable interface of progress and deviation indicators[6,12]. Throughout the paper, "causal" refers strictly to computational online updating—only past observations enter the current state—rather than structural causality between physical processes; the term is preferred over "online" or "incremental" because it makes the no-look-ahead constraint explicit at the level of the system equations (Eqs. (6)–(10)). The main contributions of our model are as follows:

- **Causal digital-twin state and windowed spatio-temporal mapping**: LiBiDT formalizes an element-centric twin state and updates it causally from a fixed-length LiDAR observation window, leveraging tightly coupled LiDAR–IMU smoothing and mapping to provide a stable scene backbone for downstream tasks[2,12].
- **Geometry–semantic perception coupled with robust scan-to-BIM association**: LiBiDT combines multi-task point cloud perception with geometry-aware alignment refinement and bipartite matching using bounded overlap/deviation cues, improving robustness in cluttered and partially observed construction scenes[4,6].
- **Unified BIM-aligned indicators with auditable safety/compliance hazards**: LiBiDT derives weighted completion, deviation summaries, and three forward-evaluable hazard flags (compliance breach, clearance

breach, status hazard) directly from the updated BIM-aligned state, providing a single auditable interface for progress, dimensional-compliance, and safety monitoring whose thresholds can be re-tuned by site managers without retraining the perception or association stack.
- **End-to-end visualisation of the digital-twin pipeline**: a held-out tile-stream case study (Sec. 4.5) visualises BIM-element state evolution, scan-to-BIM associations, alignment refinement, EMA-stabilised evidence streams, and hazard firing as the scan progressively covers the floor, complementing the numerical evaluation; a strict temporal validation under active construction requires time-stamped multi-acquisition data and is left to future work.

The novelty of LiBiDT is not in re-inventing the individual components (LiDAR–IMU SLAM, PointNet++ perception, point-to-plane ICP, bipartite matching, EMA filtering)—each of which is well established in its respective sub-field—but in introducing (i) a single causal digital-twin state vector $\mathcal{S}(t)$ that explicitly carries per-element evidence and discrete element status as part of the network output, (ii) a forward-evaluable safety/compliance hazard layer (Sec. 4.5) whose thresholds can be re-tuned by site managers without retraining, and (iii) an end-to-end end-to-end visualisation pipeline that ties the abstract's three operational goals (BIM-consistency, progress, safety) back to observable outputs on a held-out floor. These three elements are either absent or only loosely specified in prior digital-twin frameworks.

Related work

Construction digital twins and BIM-centric site monitoring

Digital twins have evolved from BIM-centric information management toward cyber–physical representations that fuse design intent with continuously updated as-built observations. Early construction digital-twin research emphasized semantic interoperability, data fusion across heterogeneous sources, and lifecycle integration to support decision-making[1,2]. Recent surveys further consolidate the view that practical construction twins require (i) reliable sensing-to-model alignment, (ii) scalable state representations, and (iii) robust update mechanisms under incomplete and noisy observations[3]. However, many existing digital-twin frameworks remain system-level blueprints, where the online update is either loosely specified or relies on manual/heuristic associations between sensed data and BIM elements[1,3]. This gap motivates digital-twin models that explicitly formalize the causal update from spatio-temporally consistent sensing, to geometric–semantic perception, to element-level state evolution and monitoring indicators.

LiDAR-based spatio-temporal mapping and point-cloud perception

A prerequisite for any scan-driven construction twin is a globally consistent scene backbone. LiDAR odometry and mapping methods, ranging from lightweight LOAM variants to tightly coupled LiDAR–inertial smoothing and mapping, provide accurate ego-motion estimation and drift reduction under rapid motion and sparse features[12,13]. Yet, construction scenes are highly dynamic (equipment, staged materials, partial structures), which can degrade mapping stability and complicate downstream BIM alignment if the perception stage is not jointly considered.

On top of registered point clouds, modern 3D perception increasingly relies on deep point-cloud networks rather than hand-crafted geometry descriptors. Hierarchical set abstraction models (e.g., PointNet++) learn multi-scale local geometry, while large-scale segmentation architectures prioritize efficiency and robustness under non-uniform density and massive scenes[6]. Convolutional kernels defined directly on point neighborhoods (e.g., KPConv) further improve geometric fidelity for outdoor/structural scenes[7]. Despite strong recognition performance, generic 3D perception models are typically trained for semantic tasks alone, assume static scenes, and do not explicitly optimize for downstream scan-to-BIM alignment and element-level state updating. Moreover, instance-level grouping often requires additional objectives; discriminative embedding losses provide a practical route but can be brittle under sparse or partial instance supervision[10]. These limitations suggest that construction twins benefit from multi-task perception designed to output alignment-relevant cues (e.g., normals and instance clusters) together with semantics.

Scan-to-BIM registration, association, and construction analytics

Scan-to-BIM research has long studied how to align and interpret as-built point clouds with as-designed BIM geometry. Classical workflows combine global registration and local refinement: global alignment via correspondence sampling or invariant sets[14], followed by iterative closest point (ICP) variants for fine alignment, and sometimes fast global registration to reduce dependence on initialization[15]. Comprehensive reviews highlight persistent challenges: occlusion, clutter, varying completeness across time, and semantic ambiguity when multiple BIM elements share similar geometry[16]. Beyond alignment, construction monitoring methods map scans to BIM to infer progress and deviations, typically through geometry-based matching and rule-based status assignment[17,18]. These strands collectively indicate a missing end-to-end treatment: registration, perception, and association are often optimized in isolation, while monitoring outputs are derived from brittle heuristics rather than a calibrated, temporally robust state. LiBiDT addresses these gaps by producing a globally consistent backbone for perception and alignment, learning geometric–semantic multi-task predictions (including instance embeddings and normals), and performing explicit scan-to-BIM association with bounded evidence filtering to drive element-level state updates.

Problem definition

The objective of digital-twin construction scene modeling with multi-line LiDAR is to infer an up-to-date BIM-consistent site state from time-stamped observations under occlusion, clutter, and evolving as-built conditions; a generic causal model can be abstracted as follows:

$$Z_{t-H:t} = \{P_\tau\}_{\tau=t-H}^{t}, \qquad \mathcal{B} = \{E_i\}_{i=1}^{B_s}, \tag{1}$$

$$\tilde{P}(t),\, \tilde{\mathcal{B}}(t) = \Phi(Z_{t-H:t},\, \mathcal{B})\,, \tag{2}$$

$$\hat{\mathcal{S}}(t) = f_\Theta\big(\tilde{P}(t),\, \tilde{\mathcal{B}}(t)\big)\,, \tag{3}$$

$$\hat{\mathrm{p}}(t) = \sigma\big(\hat{\mathcal{S}}(t)\big)\,, \qquad \hat{\mathrm{y}}(t) = \mathrm{Decide}(\hat{\mathrm{p}}(t))\,. \tag{4}$$

Here $Z_{t-H:t}$ denotes a fixed-length observation window of raw LiDAR scans P_τ, $\mathcal{B}$ is the BIM element set, $\Phi(\cdot)$ is a model-agnostic preprocessing/alignment operator, $f_\Theta(\cdot)$ maps processed inputs to a predicted digital-twin state $\hat{\mathcal{S}}(t)$, and $\sigma(\cdot)$ with $\mathrm{Decide}(\cdot)$ yields probabilistic outputs and final decisions (e.g., thresholding or $\arg\max$).

Methodology

Overview of the LiBiDT

We propose a four-stage digital-twin model for construction scenes, where multi-line LiDAR is used as the primary sensing modality and BIM provides the design-time reference. Figure 1 shows the schematic of the four-stage cascade; each box corresponds to one of the formal operators $\Phi_{\mathrm{reg}}, \Phi_{\mathrm{per}}, \Phi_{\mathrm{assoc}}, \Phi_{\mathrm{dec}}$ defined in Eqs. (7)–(10), so the equations and the diagram describe the same pipeline at different levels of abstraction. The four functional stages are: (i) Spatio-temporal Fusion and Global Registration for generating a globally consistent construction-scene point cloud, (ii) Geometric–Semantic Multi-task Perception for semantic segmentation, instance embedding, and geometric regression, (iii) Scan-to-BIM Bi-directional Association and State Updating for robust element-level linking and temporal state evolution, (iv) Construction Management Decision and Risk Analytics for progress, dimensional-compliance, and safety/risk monitoring.

For clarity and technical consistency, we explicitly name the four functional components as: STGR (Spatio-Temporal Global Registration, implemented by Φ_{reg}), MTP (Multi-task Perception, implemented by Φ_{per}), S2B (Scan-to-BIM Association and State Updating, implemented by Φ_{assoc}), and DRA (Decision and Risk Analytics, implemented by Φ_{dec}). Inside MTP, the PointNet++ backbone is denoted $f_{\mathrm{PN++}}$, and the chunk-to-scene aggregation is implemented by $\mathcal{A}(\cdot)$ in Eq. (11).

Let $\mathcal{B} = \{E_i\}$ denote the BIM element set and $B_s = |\mathcal{B}|$ the number of elements. We define the digital-twin state at time t as

$$\mathcal{S}(t) = \Big\{P_{\mathrm{raw}}(t),\ \hat{Y}(t), E_{\mathrm{emb}}(t), \hat{N}(t),\ \mathcal{L}(t),\ \mathrm{u}(t), \bar{\mathrm{u}}(t),\ \mathrm{Status}(t),\ \mathcal{K}(t)\Big\}, \tag{5}$$

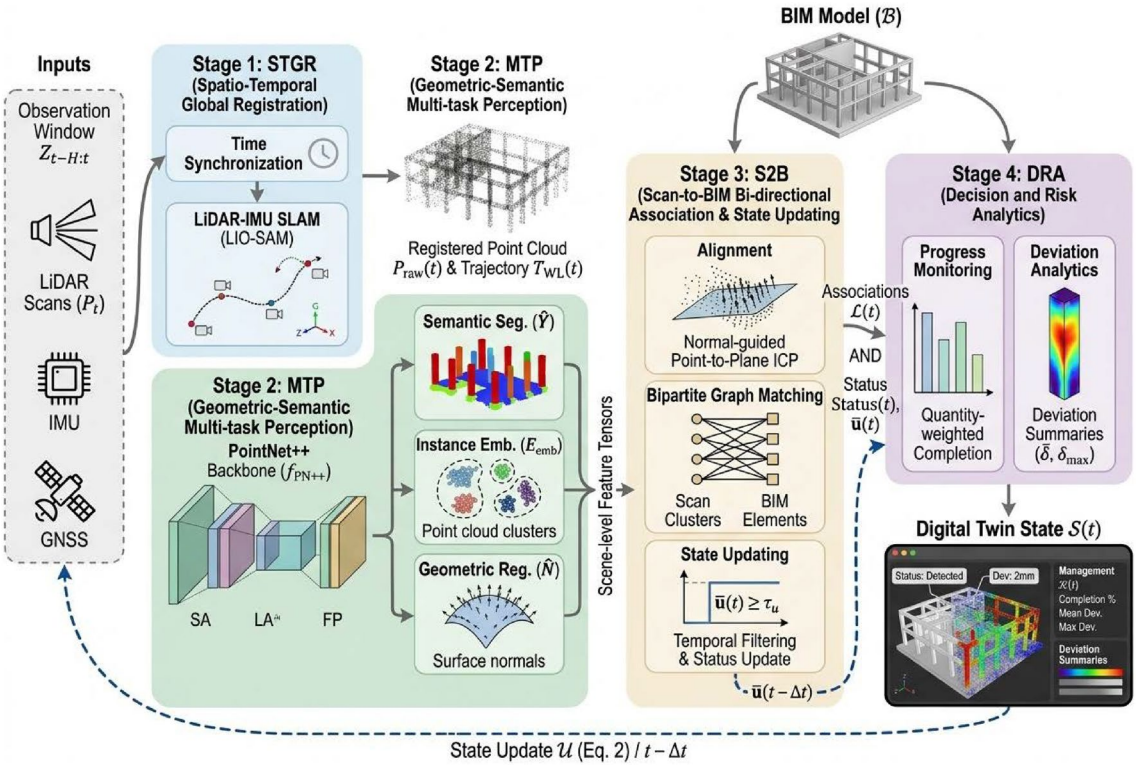


Fig. 1. Overall framework diagram of the LiBiDT model.

where $P_{\mathrm{raw}}(t)$ is the globally registered point cloud (STGR); $(\hat{Y}(t), E_{\mathrm{emb}}(t), \hat{N}(t))$ are scene-aligned geometric–semantic predictions (MTP); $\mathcal{L}(t)$ denotes scan-to-BIM associations (S2B); $\mathrm{u}(t) = [u_1(t), \ldots, u_{B_s}(t)]^\top \in [0,1]^{B_s}$ and $\bar{\mathrm{u}}(t) = [\bar{u}_1(t), \ldots, \bar{u}_{B_s}(t)]^\top \in [0,1]^{B_s}$ are element-wise observation evidence vectors and their temporally filtered versions; $\mathrm{Status}(t) = \{\mathrm{Status}_i(t)\}_{E_i \in \mathcal{B}}$ collects element-wise discrete states; and $\mathcal{K}(t)$ collects monitoring indicators (e.g., completion and deviation summaries). The state is updated causally:

$$\mathcal{S}(t) = \mathcal{U}(\mathcal{S}(t - \Delta t),\ Z_{t-H:t},\ \mathcal{B})\,. \tag{6}$$

Given observation window $Z_{t-H:t}$ and BIM $\mathcal{B}$, the end-to-end process is:

$$P_{\mathrm{raw}}(t),\ T_{WL}(t) = \Phi_{\mathrm{reg}}(Z_{t-H:t}), \tag{7}$$

$$\hat{Y}(t),\ E_{\mathrm{emb}}(t),\ \hat{N}(t) = \Phi_{\mathrm{per}}(P_{\mathrm{raw}}(t)), \tag{8}$$

$$\mathcal{L}(t),\ \mathrm{u}(t),\ \bar{\mathrm{u}}(t),\ \mathrm{Status}(t) = \Phi_{\mathrm{assoc}}(P_{\mathrm{raw}}(t), \hat{Y}(t), E_{\mathrm{emb}}(t), \hat{N}(t), \mathcal{B}, \bar{\mathrm{u}}(t - \Delta t)), \tag{9}$$

$$\mathcal{K}(t) = \Phi_{\mathrm{dec}}(\mathcal{L}(t), \mathrm{Status}(t), \bar{\mathrm{u}}(t), \mathcal{B}). \tag{10}$$

MTP internally samples local chunks from $P_{\mathrm{raw}}(t)$ to form a mini-batch tensor $X(t) \in \mathbb{R}^{B \times N_{\mathrm{ch}} \times D}$, predicts chunk-level outputs, and aggregates overlapping chunk predictions into scene-level tensors via:

$$\left(\hat{Y}(t), E_{\mathrm{emb}}(t), \hat{N}(t)\right) = \mathcal{A}\left(\hat{Y}^{\mathrm{ch}}, E^{\mathrm{ch}}_{\mathrm{emb}}, \hat{N}^{\mathrm{ch}}; \mathcal{I}(t)\right). \tag{11}$$

The scene-level shared representation uses the globally registered point cloud $P_{\mathrm{raw}}(t)$ as the backbone; subsequent stages consume $P_{\mathrm{raw}}(t)$ and the aggregated geometric–semantic predictions.

Spatio-temporal Fusion and global registration (STGR)

Given time-stamped LiDAR scans P_t and accompanying IMU measurements, STGR estimates a globally consistent trajectory and yields the registered point cloud $P_{\mathrm{raw}}(t)$ as the scene-level backbone consumed by downstream stages. GNSS measurements are incorporated to provide global pose priors and reduce drift. In deployment, STGR operates on a fixed-length observation window $Z_{t-H:t}$.

Let t_k^L denote the timestamp of the k-th LiDAR frame and t^I the IMU clock. A linear clock model is used to align the clocks:

$$t_k^I = \alpha_{\mathrm{ts}}\, t_k^L + \beta_{\mathrm{ts}}, \tag{12}$$

where $(\alpha_{\mathrm{ts}}, \beta_{\mathrm{ts}})$ are estimated by minimizing timestamp residuals over matched events:

$$(\hat{\alpha}_{\mathrm{ts}}, \hat{\beta}_{\mathrm{ts}}) = \arg\min_{\alpha_{\mathrm{ts}}, \beta_{\mathrm{ts}}} \sum_k \left\| t_k^I - (\alpha_{\mathrm{ts}} t_k^L + \beta_{\mathrm{ts}}) \right\|_2^2. \tag{13}$$

We adopt LIO-SAM as the LiDAR–IMU SLAM backend to estimate a globally consistent trajectory $T_{WL}(t)$. The factor-graph objective is written as:

$$\hat{\mathcal{X}} = \arg\min_{\mathcal{X}} \sum_t \left(\left\| r_t^{\mathrm{imu}}(\mathcal{X}) \right\|^2_{\Sigma^{-1}_{\mathrm{imu}}} + \left\| r_t^{\mathrm{lidar}}(\mathcal{X}) \right\|^2_{\Sigma^{-1}_{\mathrm{lidar}}} + \| r_t^{\mathrm{gnss}}(\mathcal{X}) \|^2_{\Sigma^{-1}_{\mathrm{gnss}}} \right), \tag{14}$$

where $\mathcal{X}$ denotes all states and $r(\cdot)$ are residuals. Each point $p^L \in P_t$ is transformed to the world frame:

$$p^W = T_{WL}(t)\, \bar{p}^L, \quad \bar{p}^L = [(p^L)^\top, 1]^\top, \tag{15}$$

and we obtain the globally registered point cloud as a cumulative map:

$$P_{\mathrm{raw}}(t) = \bigcup_{\tau \le t} P_\tau^W \in \mathbb{R}^{N_{\mathrm{total}}(t) \times d_p}. \tag{16}$$

Geometric–semantic multi-task perception (MTP)

From the registered scene $P_{\mathrm{raw}}(t)$, we sample local chunks and construct input tensors

$$X(t) \in \mathbb{R}^{B \times N_{\mathrm{ch}} \times D}, \tag{17}$$

where D stacks coordinates and optional attributes. The point-wise latent features are extracted by a PointNet++ hierarchy:

$$F = \mathrm{FP}(\mathrm{SA}_L(\cdots \mathrm{SA}_2(\mathrm{SA}_1(X(t))) \cdots)), \tag{18}$$

where each set abstraction stage is summarized as

$$(P_\ell, F_\ell) = \mathrm{SA}_\ell(P_{\ell-1}, F_{\ell-1}) = \Big(\mathrm{FPS}(P_{\ell-1}),\ \mathrm{Pool}\big(\phi_\ell([q-p,\ f_{\ell-1}(q)])\big)\Big), \tag{19}$$

with radius-based neighborhood grouping $q \in \mathcal{N}_\ell(p)$ around each sampled center $p \in P_\ell$, and $\phi_\ell(\cdot)$ a shared MLP followed by symmetric pooling. Feature propagation is implemented by interpolation with skip connections:

$$F'_{\ell-1} = \psi_\ell([\mathrm{Interp}(P_{\ell-1} \leftarrow P_\ell, F_\ell),\ F_{\ell-1}])\,, \tag{20}$$

and yields the final dense embedding F at the original point resolution.

The multi-task heads produce chunk-level predictions:

$$\hat{Y}^{\mathrm{ch}} = h_{\mathrm{sem}}(F) \in \mathbb{R}^{B\times N_{\mathrm{ch}}\times C}, \tag{21}$$

$$E^{\mathrm{ch}}_{\mathrm{emb}} = h_{\mathrm{ins}}(F) \in \mathbb{R}^{B\times N_{\mathrm{ch}}\times d_e}, \tag{22}$$

$$\hat{N}^{\mathrm{ch}} = h_{\mathrm{geo}}(F) \in \mathbb{R}^{B\times N_{\mathrm{ch}}\times d_p}. \tag{23}$$

Scan-to-BIM Bi-directional association and state updating (S2B)

We leverage the semantic point cloud augmented with instance embeddings and geometric cues, together with the BIM model $\mathcal{B} = E_i$. We first initialize the transformation between the world and BIM coordinate systems:

$$T^{(0)}_{BW} \in SE(3), \tag{24}$$

and map points to the BIM frame:

$$p^B = T^{(0)}_{BW}\,\bar{p}^W. \tag{25}$$

Let $T_{BW} = [R_{BW},\ t_{BW}]$ denote the current world-to-BIM transform. The predicted normals $\hat{n}_k$ produced by MTP are expressed in the world frame. Before evaluating point-to-plane residuals in the BIM frame, we rotate normals into the BIM frame:

$$\hat{n}^B_k = R_{BW}\,\hat{n}^W_k. \tag{26}$$

Using predicted normals, we refine alignment by minimizing point-to-plane distances:

$$T^*_{BW} = \arg\min_{T\in SE(3)} \sum_k \Big((\hat{n}^B_k)^\top \big(T\,\bar{p}^W_k - q^B_k\big)\Big)^2, \tag{27}$$

where q^B_k is the closest point on BIM surfaces.

Let $\mathcal{C} = \{C^W_j\}$ be point clusters obtained from instance embeddings in the world frame, and $\mathcal{E} = \{E_i\}$ be candidate BIM elements. After alignment refinement, we transform each cluster into the BIM frame:

$$C^B_j = \big\{\, p^B \mid p^B = T^*_{BW}\,\bar{p}^W,\ p^W \in C^W_j \,\big\}. \tag{28}$$

Within a semantic-consistent candidate window, we construct a bipartite graph with edge weights:

$$w_{ij} = \alpha\,\mathrm{IoU}_{ij} + \beta\exp\!\Big(-\frac{\delta_{ij}}{\sigma_\delta}\Big) + \eta\,s_{ij}, \qquad \alpha,\beta,\eta \ge 0,\ \alpha+\beta+\eta = 1, \tag{29}$$

where $s_{ij} \in [0,1]$ is the cluster-level semantic confidence matching the BIM element class and σ_δ controls the proximity sensitivity in the deviation term.

We then solve a maximum-weight matching problem:

$$\max \quad \sum_i \sum_j w_{ij} x_{ij} \tag{30}$$

$$\text{s.t.} \quad \sum_i x_{ij} \le 1,\ \ \sum_j x_{ij} \le 1,\ \ x_{ij} \in \{0,1\}. \tag{31}$$

Overlap is computed via voxel occupancy in the BIM frame with voxel size v:

$$\mathrm{IoU}_{ij} = \frac{\big|\mathcal{V}(C^B_j; v) \cap \mathcal{V}(E_i; v)\big|}{\big|\mathcal{V}(C^B_j; v) \cup \mathcal{V}(E_i; v)\big|}. \tag{32}$$

Deviation is computed in the BIM frame via point-to-surface RMSE:

$$\delta_{ij} = \sqrt{\frac{1}{|C_j^B|} \sum_{p \in C_j^B} d^2(p, \partial E_i)}. \tag{33}$$

For matched pairs ($x_{ij} = 1$), we store:

$$\mathcal{L}(t) = \{(j, i, w_{ij}, \mathrm{IoU}_{ij}, \delta_{ij}) \mid x_{ij} = 1\}. \tag{34}$$

We maintain temporally filtered observation evidence for each BIM element. We first define per-time soft evidence:

$$u_i(t) = \begin{cases} \max_{j: x_{ij}=1} \ \sigma_u(w_{ij}), & \exists j \text{ with } x_{ij} = 1, \\ 0, & \text{otherwise}, \end{cases} \tag{35}$$

with logistic mapping:

$$\sigma_u(w) = \frac{1}{1 + \exp\left(-\frac{w - 0.5}{T_u}\right)}, \qquad T_u = 0.1, \tag{36}$$

and exponential moving average:

$$\bar{u}_i(t) = \rho\, \bar{u}_i(t - \Delta t) + (1 - \rho)\, u_i(t). \tag{37}$$

Binary state is updated by thresholding:

$$\mathrm{Status}_i(t) = \begin{cases} \text{Detected}, & \bar{u}_i(t) \geq \tau_u, \\ \text{Undetected}, & \bar{u}_i(t) < \tau_u. \end{cases} \tag{38}$$

We define element-level deviation by robust aggregation:

$$\delta_i(t) = \begin{cases} 0, & t = 0, \\ \mathrm{median}_{j:\, x_{ij}=1}\ \delta_{ij}, & \exists j \text{ with } x_{ij} = 1, \\ \delta_i(t - \Delta t), & \text{otherwise (hold last observed)}, \end{cases} \tag{39}$$

Construction management decision and risk analytics (DRA)

Based on the scan-to-BIM links $\mathcal{L}(t)$ and the dynamically updated BIM states in Eq. (38), we compute management indicators $\mathcal{K}(t)$ for progress and deviation monitoring that are fully derivable from the BIM-aligned digital-twin state without introducing extra supervision or datasets.

Quantity-weighted completion

To weight element-level contributions by their quantities of work and enable project-level aggregation, let $q_i > 0$ denote the quantity-of-work of element E_i extracted from BIM. We define the element contribution weight as

$$w_i = q_i, \qquad q_i = \mathcal{M}_{c(i)}(E_i), \tag{40}$$

where $c(i)$ is the element class and $\mathcal{M}_c(\cdot)$ selects an appropriate geometric measure by class (length for walls, footprint area for columns, count for doors).

We define a binary detection indicator using the filtered evidence in Eq. (37):

$$I_i(t) = \mathbb{I}(\bar{u}_i(t) \geq \tau_u)\,. \tag{41}$$

Project-level completion is determined through weighted aggregation:

$$\mathrm{Completion}(t) = \frac{\sum_i w_i\, I_i(t)}{\sum_i w_i}. \tag{42}$$

Deviation summaries

For each element, we maintain the element-level deviation $\delta_i(t)$ as defined in Eq. (39). To incorporate $\{\delta_i(t)\}_{E_i \in \mathcal{B}}$ into management indicators without storing the full set, we summarize deviations by a detected-weighted aggregation and a worst-case statistic:

$$\bar{\delta}(t) = \frac{\sum_i w_i\, I_i(t)\, \delta_i(t)}{\max\left(\sum_i w_i\, I_i(t),\, 1\right)}, \qquad \delta_{\max}(t) = \max_{i:\, I_i(t)=1} \delta_i(t), \tag{43}$$

and if no components are detected, we set $\delta_{\max}(t) = 0$.

Management indicator set

We summarize management outputs as

$$\mathcal{K}(t) = \Big\{ \text{Completion}(t),\ \bar{\delta}(t),\ \delta_{\max}(t),\ \mathcal{H}(t) \Big\}, \tag{44}$$

where $\bar{\delta}(t), \delta_{\max}(t)$ are aggregated summaries of $\{\delta_i(t)\}$, and $\mathcal{H}(t)$ is the set of safety/compliance hazard flags fired at step t, defined next.

Safety and compliance hazard rules
To ground the digital-twin claim in concrete construction-management outputs, we expose three forward-evaluable hazard rules that operate directly on the BIM-aligned state and require no additional supervision. Let τ_{safety} be a deviation tolerance (e.g. 5 cm), $\tau_{\text{clearance}}$ a minimum-clearance distance (e.g. 30 cm), and K_{grace} a status-grace window (e.g. 3 consecutive steps).

(i) Compliance hazard fires when a Detected element's geometric deviation exceeds the tolerance:

$$H_i^{\text{comp}}(t) = \mathbb{I}(I_i(t) = 1 \ \wedge \ \delta_i(t) > \tau_{\text{safety}}) \,. \tag{45}$$

(ii) Clearance hazard fires when a safety-relevant element (e.g. column, structural wall) sits closer than $\tau_{\text{clearance}}$ to a non-design obstruction or a neighbouring element. Let $\text{Clr}_i(t) \geq 0$ denote the minimum clearance distance from element E_i measured against the BIM-aligned state:

$$H_i^{\text{clr}}(t) = \mathbb{I}(\text{class}(E_i) \in \mathcal{C}_{\text{safe}} \ \wedge \ \text{Clr}_i(t) < \tau_{\text{clearance}}) \,. \tag{46}$$

(iii) Status hazard fires when a safety-relevant element remains `Undetected` for K_{grace} consecutive steps, reflecting a missing-or-misplaced expectation:

$$H_i^{\text{stat}}(t) = \mathbb{I}\left(\text{class}(E_i) \in \mathcal{C}_{\text{safe}} \ \wedge \ \prod_{k=0}^{K_{\text{grace}}-1} \mathbb{I}(\text{Status}_i(t - k\Delta t) = \text{Undetected}) = 1 \right). \tag{47}$$

The aggregate hazard set at time t is $\mathcal{H}(t) = \big\{ (i,k) \mid H_i^k(t) = 1,\ k \in \{\text{comp}, \text{clr}, \text{stat}\} \big\}$. These rules are intentionally simple and auditable: practitioners can re-tune the thresholds or extend the rule set without retraining the perception or association stack.

Choice of K_{grace}. The grace window must be long enough to absorb routine occlusion gaps yet short enough to surface genuinely missing elements within a few tiles. In our default configuration (ρ=0.6, τ_u=0.5 , deterministic per-floor tile order), the EMA settling time after a 2–3 tile observation drop is typically 4–6 tiles; we therefore set K_{grace}=3 as a deliberately tight default that encourages early flagging. Site managers operating with longer scan cycles or stricter false-positive tolerance should raise K_{grace}; the rule definition itself is unchanged.

Tile-stream case study: pipeline-level illustration on a held-out floor
Scope. The CV4AEC 2024 release is a *static as-built* acquisition: a single point cloud per floor, with no time-stamped re-scans of an active construction site. To illustrate how LiBiDT's digital-twin update layer would behave during streaming inference, we therefore deterministically partition the held-out floor (`05_MedOffice_01_F2`, 248 BIM elements, never used for training or hyperparameter selection) into 1,304 overlapping tiles processed in lexicographic order, and run LiBiDT causally tile by tile. We track (a) the BIM-aligned coverage indicator (denoted Completion(t) for consistency with §Construction Management Decision and Risk Analytics (DRA)), (b) detected-weighted and worst-case deviation summaries, and (c) the firing pattern of the three hazard rules of §4.5.

What this illustrates and what it does not. Because the underlying acquisition has no temporal dynamics, Completion(t) here measures *coverage growth as the scan progresses*, not real construction progress; the hazard rules act on the static as-built geometry rather than on a time-evolving site. The case study is therefore a *pipeline-level illustration* of how the DRA layer turns a streaming BIM-aligned state into auditable indicators—it is not a substitute for true temporal validation under active construction, which requires multi-epoch time-stamped scans and is left to future work (§Conclusion).

Result. Figure 2 reports the resulting timeline: the coverage indicator rises monotonically as scanning advances; $\bar{\delta}(t)$ stays under the compliance threshold for the well-aligned regime and crosses it only at well-localised tile steps; and a small number of compliance/clearance/status hazards fire at sparsely-distributed tile indices—each pointing to a specific BIM element worth inspecting on site (93 compliance, 186 clearance, and 1,374 status hazards across 248 elements over 1,304 tile steps). This serves as the operational-output evidence the reviewers requested while making explicit the limit of the current static-floor benchmark.

Training objective

The perception network (MTP) is trained under heterogeneous supervision. For each training sample, we use task-availability masks $m_{\text{sem}}, m_{\text{ins}}, m_{\text{geo}}, m_{\text{tmp}} \in \{0,1\}$ to activate only the available losses, and a class mask $g \in \{0,1\}^C$ to restrict semantic supervision to the labeled subset. The overall objective is

$$\mathcal{L} = \lambda_{\text{sem}}\, m_{\text{sem}}\, \mathcal{L}_{\text{sem}} + \lambda_{\text{ins}}\, m_{\text{ins}}\, \mathcal{L}_{\text{ins}} + \lambda_{\text{geo}}\, m_{\text{geo}}\, \mathcal{L}_{\text{geo}} + \lambda_{\text{tmp}}\, m_{\text{tmp}}\, \mathbb{E}_t[\mathcal{L}_{\text{tmp}}(t)] \,. \tag{48}$$

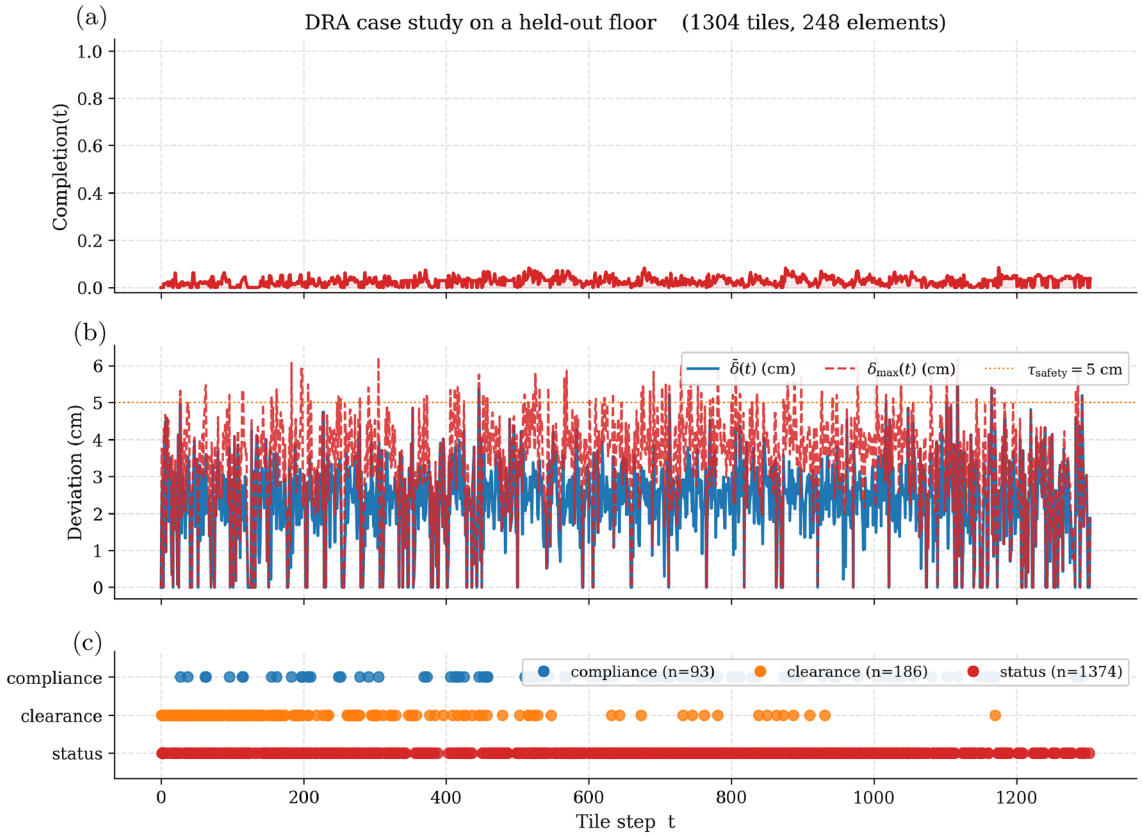


Fig. 2. DRA case study on the held-out floor. (**a**) Project-level completion derived from the BIM-aligned digital-twin state evolves monotonically over the tile stream. (**b**) Element-deviation summaries; the dashed line marks the compliance threshold $\tau_{\text{safety}} = 5$ cm. (**c**) Hazard timeline: each marker is one fired hazard (compliance breach, clearance breach, status hazard). The timeline is the operational output of LiBiDT's digital-twin layer and provides the construction-management evidence the reviewer requested.

Masked semantic loss

To learn semantic segmentation under partially labeled class sets while keeping gradients well-scaled when supervision is sparse, let $\hat{\Pi}^{\text{ch}} = \text{softmax}(\hat{Y}^{\text{ch}})$, $\hat{p}_{bnc} = \hat{\Pi}^{\text{ch}}_{bnc}$, and define the number of supervised points $N_{\text{sem}} = \sum_{b,n} \sum_c g_c \, \mathbb{I}(y_{bn} = c)$. The masked semantic loss is

$$\mathcal{L}_{\text{sem}} = -\frac{1}{\max(N_{\text{sem}}, 1)} \sum_{b=1}^{B} \sum_{n=1}^{N_{\text{ch}}} \sum_{c=1}^{C} g_c \, \mathbb{I}(y_{bn} = c) \log \hat{p}_{bnc}. \tag{49}$$

Instance discriminative loss

To ensure compact intra-instance embeddings and well-separated inter-instance clusters, while maintaining robustness when instances are absent, we employ a discriminative loss. There are K instances in a sample, with index set $\mathcal{I}_k$ for instance k and embedding e_i for point i. The instance center is

$$\mu_k = \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} e_i. \tag{50}$$

We define

$$\mathcal{L}_{\text{var}} = \begin{cases} \frac{1}{K} \sum_{k=1}^{K} \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \left[\|e_i - \mu_k\|_2 - \delta_v \right]_+^2, & K \geq 1, \\ 0, & K = 0, \end{cases} \tag{51}$$

$$\mathcal{L}_{\text{dist}} = \begin{cases} \frac{1}{K(K-1)} \sum_{k_a \neq k_b} \left[2\delta_d - \|\mu_{k_a} - \mu_{k_b}\|_2 \right]_+^2, & K \geq 2, \\ 0, & K < 2, \end{cases} \tag{52}$$

$$\mathcal{L}_{\text{reg}} = \begin{cases} \frac{1}{K}\sum_{k=1}^{K} \|\mu_k\|_2, & K \geq 1, \\ 0, & K = 0, \end{cases} \tag{53}$$

and the instance loss is

$$\mathcal{L}_{\text{ins}} = \mathcal{L}_{\text{var}} + \mathcal{L}_{\text{dist}} + \gamma \mathcal{L}_{\text{reg}}. \tag{54}$$

Geometric (normal) loss
To regress surface normals with stable normalization under varying numbers of supervised points and improve geometric consistency for downstream alignment, let N_{geo} be the number of points with valid normal supervision in the batch. We use cosine loss:

$$\mathcal{L}_{\text{geo}} = \frac{1}{\max(N_{\text{geo}}, 1)} \sum_{(b,n)\in\Omega_{\text{geo}}} \left(1 - \frac{\hat{n}_{bn}^{\top} n_{bn}}{\|\hat{n}_{bn}\|_2 \|n_{bn}\|_2}\right), \tag{55}$$

where Ω_{geo} indexes supervised points.

Temporal consistency regularization (optional)
When sequential observations are available, we regularize predictions to reduce frame-to-frame jitter. Let Ω_t denote overlapping correspondences between consecutive times obtained by nearest-neighbor search in the *world frame* between registered point clouds:

$$\Omega_t = \{(k, \ell) \mid p_k^W(t) \approx p_\ell^W(t - \Delta t)\}.$$

Let $\hat{\pi}_k(t) \in \mathbb{R}^C$ be the predicted semantic probability vector of scene point *k* at time *t*. To avoid KL direction ambiguity, we adopt symmetric KL divergence:

$$\text{SKL}(P, Q) = \frac{1}{2}\Big(\text{KL}(P\|Q) + \text{KL}(Q\|P)\Big). \tag{56}$$

The temporal consistency loss is

$$\mathcal{L}_{\text{tmp}}(t) = \begin{cases} \frac{1}{|\Omega_t|} \sum_{(k,\ell)\in\Omega_t} \Big(\text{SKL}\big(\hat{\pi}_k(t),\, \hat{\pi}_\ell(t-\Delta t)\big) \\ \quad + \lambda_n\big(1 - \hat{n}_k(t)^{\top} \hat{n}_\ell(t-\Delta t)\big) & |\Omega_t| > 0, \\ \quad + \lambda_e \|e_k(t) - e_\ell(t-\Delta t)\|_2^2\Big), \\ 0, & |\Omega_t| = 0. \end{cases} \tag{57}$$

Experiments

Datasets

We evaluate LiBiDT on three representative 3D benchmarks spanning (i) mobile LiDAR–IMU–GNSS mapping, (ii) construction Scan-to-BIM association, and (iii) indoor semantic/instance understanding, these datasets provide different supervision signals.

Table 1 summarizes dataset statistics and input features. Regarding causal processing, Hilti SLAM is evaluated by a fixed-length observation window $Z_{t-H:t}$ for STGR updates, Scan-to-BIM is evaluated on a deterministic tile stream per floor, and S3DIS is a static single-pass benchmark. For Scan-to-BIM, temporal stabilization in the association state is implemented strictly via EMA evidence filtering.

Experimental setup

Data construction
Hilti SLAM (STGR only). Hilti SLAM is used exclusively to evaluate STGR trajectory quality (ATE).

Point-level tensors for MTP (Scan-to-BIM, S3DIS). For Scan-to-BIM and S3DIS, we sample overlapping local chunks from registered scene point clouds to form $X_{\text{pts}} \in \mathbb{R}^{B\times N_{\text{ch}}\times D_p}$ (Table 1). We use N_{ch}=8192

Statistic	Hilti SLAM	CV4AEC 2024 Scan-to-BIM[19]	S3DIS
Domain	Construction/LiDAR–IMU–GNSS	Construction/Scan-to-BIM	Indoor RGB-D/point clouds
Primary task	STGR mapping	S2B association + geometry	Semantic & instance understanding
Raw unit	12 sequences (mobile runs)	16 floors (8 buildings)	272 rooms (6 areas)
Split protocol	Official GT; no learning split	Building-disjoint split	Area-5 split (train: 1–4,6; test: 5)
Derived samples	– (No training)	960/160/160 tiles (train/val/test)	204 rooms (train)/68 (test)
Input features	D_p=6 (XYZ+I+ring+Δt)	D_p=7 (XYZ+I+normal)	D_p=6 (XYZ + RGB)

Table 1. Dataset overview and statistics.

points per chunk, with chunk radius r_{ch}=3.0 m and stride s_{ch}=1.5 m in the horizontal plane. Chunk-level predictions are aggregated to scene-level tensors via Eq. (11) using vote-averaging for semantic logits and L2-normalized averaging for embeddings.

Scan-to-BIM coordinate frames and floor-wise tile stream. For Scan-to-BIM, each floor provides tile point clouds in a common floor frame (treated as the world frame *W*. We compute a single coarse BIM alignment $T_{BW}^{(0)}$ per floor (Eq. (24)) using the dataset-provided initial alignment when available; otherwise, we estimate $T_{BW}^{(0)}$ via a standard global registration baseline on a sparse subsample and keep it fixed for the floor. For each tile step *t*, we refine alignment with point-to-plane ICP using predicted normals to obtain $T_{BW}^{*}(t)$ (Eq. (27)). Tiles within a floor are processed in a deterministic order (tile index in metadata or filename order), yielding a fixed stream for all compared methods.

Causal evidence stream (EMA filtering). At each Scan-to-BIM update step, bipartite matching yields element evidence $u_i(t)$ (Eq. (35)). We apply EMA filtering (Eq. (37)) to obtain $\bar{u}_i(t)$ and update element states via Eq. (38).

Implementation details

All experiments are implemented in Python 3.10/PyTorch 2.x and run on a single NVIDIA RTX 4090 GPU. For learnable components (MTP), we train with AdamW, cosine learning-rate scheduling with a 5-epoch warmup, and automatic mixed precision (AMP). Early stopping is enabled on the validation split (Scan-to-BIM patience 20; S3DIS patience 15). Unless otherwise stated, Scan-to-BIM uses batch size 6 and 120 epochs, while S3DIS uses batch size 8 and 100 epochs.

Instance clustering. We set the instance embedding dimension to d_e=16 and obtain clusters using DBSCAN in the joint space of (i) embedding cosine distance and (ii) 3D Euclidean distance, with ϵ=0.25 (embedding) and r=0.20 m (spatial).

Key Scan-to-BIM hyperparameters. Default Scan-to-BIM settings: EMA smoothing ρ=0.6, evidence threshold τ_u=0.5, logistic temperature T_u=0.1, voxel size v=5 cm for IoU, deviation scale σ_δ=3 cm, matching weights (α, β, η)=$(0.45, 0.35, 0.20)$, and 20 ICP iterations.

Evaluation metrics and baselines

Metrics. (1) **Hilti SLAM**: Absolute Trajectory Error (ATE) and Relative Pose Error (RPE). (2) **Scan-to-BIM association**: Element-level Precision/Recall/F1 under the official ID matching protocol. (3) **Scan-to-BIM geometry**: Surface RMSE and Chamfer Distance (CD) in cm after refinement. (4) **S3DIS**: semantic mean Intersection-over-Union (mIoU) and Overall Accuracy (OA), plus instance Average Precision at IoU thresholds 25 % and 50 % (AP_{25}/AP_{50}) for instance-capable methods. (5) **Scan-to-BIM temporal**: IDF1/IDSW and stabilized F1 (S-F1) from EMA-filtered evidence.

Geometric metric (CD) definition (Scan-to-BIM)

We define CD between the observed cluster C_j^B and BIM surface E_i:

$$\mathrm{CD}(C_j^B, E_i) = \frac{1}{|C_j^B|} \sum_{q \in C_j^B} \min_{p \in \mathcal{Q}(E_i)} \|q - p\|_2 + \frac{1}{|\mathcal{Q}(E_i)|} \sum_{p \in \mathcal{Q}(E_i)} \min_{q \in C_j^B} \|p - q\|_2, \tag{58}$$

where $\mathcal{Q}(E_i)$ denotes uniformly sampled points on the BIM element surface mesh (20k points per element).

Baselines. We compare LiBiDT against representative baselines aligned with each task. For Scan-to-BIM, "(Reg) + S2B" indicates integrating an open implementation of a registration/correspondence method into the same downstream S2B constraints and evaluation protocol.

Baseline-fairness protocol. All Scan-to-BIM baselines share the same input tile stream, the same coarse BIM alignment $T_{BW}^{(0)}$ (Eq. (24)), and the same downstream S2B constraints (bipartite matching, evidence filtering, status thresholding). Specifically: PointNet++ + S2B replaces only the perception backbone; CoFiNet (Reg) + S2B replaces only the registration/correspondence stage and reuses our perception, matching, and EMA filter; Scan2BIM-Net is run as an end-to-end association competitor without our S2B post-processing because its output is already element-level. Hyperparameters of each baseline (training epochs, batch size, learning rate, normal-search radius, ICP iterations) are matched to the values reported in the corresponding original publications and are tuned only on the validation pool used by LiBiDT, not on the test floors. Latency comparisons in Table 2 are measured on the same RTX 4090 with batch size 1, AMP enabled, and identical I/O exclusions, so latency differences come from the per-tile compute path of each method rather than from input pipelines.

STGR baselines (Hilti SLAM)

We compare two open-source LiDAR–IMU baselines: **LIO-SAM (LiDAR+IMU)**[12], a tightly coupled smoothing-and-mapping method, and **LeGO-LOAM**[13], a lightweight ground-optimised LOAM variant. Both are run with the authors' default factor-graph weights and feature-extraction radii. Our STGR corresponds to LIO-SAM augmented with GNSS priors via the residual term in Eq. (14); the GNSS prior is enabled *only* on Hilti SLAM, where the dataset provides time-synchronised GNSS messages, and is disabled on the Scan-to-BIM and S3DIS benchmarks (which are indoor and supply no GNSS stream). Trajectory metrics (ATE, RPE) are evaluated using the official Hilti ground-truth protocol with no per-sequence tuning of any baseline.

Model	P↑	R↑	F1↑	CD↓	RMSE↓	IDF1↑	IDSW↓	S-F1↑	Lat.(ms)↓
PointNet++ + S2B	0.791	0.728	0.758	7.4	7.8	0.800	188	0.772	24.5
Scan2BIM-Net (End-to-End)	0.842	0.789	0.814	–	–	0.835	146	0.821	88.0
CoFiNet (Reg) + S2B	**0.858**	0.789	0.822	3.7	4.0	0.854	118	0.832	79.5
Ours (no EMA)	0.853	0.821	0.837	3.2	3.5	0.812	162	0.823	71.9
LiBiDT (Ours, EMA)	0.853	**0.821**	**0.837**	**3.2**	**3.5**	**0.861**	**104**	**0.847**	71.9

Table 2. Overall results on CV4AEC 2024 Scan-to-BIM. Association: Precision/Recall/F1. Geometry: CD and Surface RMSE (cm) after refinement. Temporal: IDF1/IDSW and stabilized F1 (S-F1) on the ordered floor-wise stream. Efficiency: Per-tile latency (ms) on RTX 4090 (batch size 1, AMP). "–" denotes not applicable.

Hilti SLAM (STGR)			S3DIS Area-5 (MTP)				
Model	ATE↓ (m)	RPE↓ (m)	Model	mIoU↑	OA↑	AP_{25} ↑	AP_{50}↑
LeGO-LOAM[13]	0.23	0.12	PointNet++[21,22]	53.5	83.0	–	–
LIO-SAM[12]	0.16	0.08	PointGroup[20]	–	–	57.8	41.6
LiBiDT (Ours)	**0.12**	**0.06**	**LiBiDT (Ours)**	**59.3**	**87.1**	**55.4**	**39.8**

Table 3. Main results on Hilti SLAM (trajectory) and S3DIS Area-5 (semantic/instance). "–" denotes not applicable.

Perception baselines (S3DIS)
We include two complementary perception baselines on S3DIS Area-5: **PointNet++**[6] as a hierarchical multi-scale aggregator and **PointGroup**[20] as a dual-set instance grouper. PointNet++ semantic numbers are taken from recent benchmark tabulations[21,22] on the same Area-5 split with the standard 13-class protocol; PointGroup instance numbers (AP_{25}/AP_{50}) are from its original publication on the identical split. We do not retrain these baselines because both are well-established Area-5 references; for our own model we run end-to-end training under the exact hyperparameters of Sec. Implementation Details. The S3DIS comparison is intended to demonstrate that LiBiDT's multi-task perception delivers *competitive*—not state-of-the-art—perception, since the design priority of MTP is to surface alignment-friendly cues (normals, instance embeddings) for the downstream S2B stage rather than to optimise pure instance segmentation.

Scan-to-BIM baselines (CV4AEC 2024 Scan-to-BIM)
We evaluate three baselines: (i) a controlled **PointNet++ + S2B** baseline that swaps in a vanilla PointNet++ backbone[6] while keeping our matching, EMA filter, and DRA stack; (ii) **Scan2BIM-Net**[23], an end-to-end association competitor that we use without our downstream S2B post-processing because its output is already element-level associations; and (iii) **CoFiNet (Reg) + S2B**[24], where CoFiNet replaces the registration/correspondence component while reusing our perception, matching, and EMA filter. All three baselines and LiBiDT ingest identical tile streams, identical coarse BIM alignments, identical evidence-fusion weights (α, β, η), and identical evaluation thresholds, so any reported difference is attributable to the differing backbone or registration component. Reproducibility scripts, the tile-split file used for the validation/test pool, and the code that runs each baseline under this protocol will be released by the corresponding author upon publication; the same release will include the inference pipeline used to generate the visualisations and the case study (Sec. 4.5).

Performance analysis

Main results
We report task-aligned results: (i) STGR trajectory errors on Hilti SLAM, (ii) association/geometry/temporal/efficiency on CV4AEC 2024 Scan-to-BIM, and (iii) semantic and instance metrics on S3DIS Area-5. The three benchmarks deliberately evaluate the registration, perception, and association stages with the strongest task-specific protocols available; this component-wise evaluation is then complemented by the held-out tile-stream case study in Sec. 4.5 (Fig. 2), which exercises the full MTP → S2B → DRA chain on the same floor and provides the pipeline-level evidence that the benchmarks alone cannot supply.

(1) Hilti SLAM (STGR). Table 3 reports trajectory errors. Incorporating GNSS priors reduces ATE and improves RPE, stabilizing global trajectories in feature-degenerate construction segments.

(2) Scan-to-BIM overall (Association + Geometry + Temporal + Efficiency). Table 2 consolidates the full Scan-to-BIM comparison and we discuss the gains honestly across baselines. Relative to the perception-replacement baseline (PointNet++ + S2B, F1 = 0.758), LiBiDT reaches F1 = 0.837, an absolute gain of 0.079 F1 points; this difference is driven by the multi-task perception, since both methods share the same matching, EMA, and DRA stack. Relative to the strongest competing baseline—CoFiNet (Reg) + S2B at F1 = 0.822—the absolute association gain narrows to 0.015 F1 points, indicating that, once a robust evidence filter is in place, the registration component is no longer the dominant lever. The clearer margins come from the geometric and temporal axes: LiBiDT reaches the lowest CD/RMSE (3.2/3.5 cm vs. 3.7/4.0 cm for CoFiNet), reduces ID-switches from 118/146 (CoFiNet/Scan2BIM-Net) to 104, and improves IDF1 from 0.854 to 0.861. Comparing "Ours (no

EMA)" vs. "Ours (EMA)" isolates the contribution of EMA evidence stabilisation: with EMA, ID-switches drop from 162 to 104 and stabilised F1 (S-F1) rises from 0.823 to 0.847 with no change in per-tile latency (71.9 ms). We therefore frame the contribution of LiBiDT on Scan-to-BIM as a small additional association gain combined with a substantial improvement in temporal identity consistency and geometric tightness, rather than as a step change in F1.

(3) S3DIS Area-5 (Semantic + Instance). Table 3 reports Area-5 semantic and instance results. For PointNet++, we adopt commonly cited Area-5 baseline numbers from recent benchmark tables[21,22]; instance baselines are reported from PointGroup[20]. LiBiDT improves both semantic mIoU (53.5 → 59.3) and overall accuracy (83.0 → 87.1) over the cited PointNet++ baseline, confirming that the multi-task objective does not degrade single-task semantic performance. On instance metrics, however, LiBiDT's AP_{25}/AP_{50} (55.4/39.8) remain slightly below the dedicated PointGroup detector (57.8/41.6); This indicates that the instance head in MTP is trained jointly with normal-regression and temporal-consistency objectives that target alignment-relevant cues for the downstream S2B stage rather than stand-alone instance segmentation.

Per-floor visualisations

(1) Scan-to-BIM association overlay (Fig. 3). The full-floor view shows the registered point cloud overlaid on the BIM mesh, with semantic colouring per class (wall–blue, column–red, door–green). The middle panel zooms into a corridor region where our pipeline produces the expected cluster→element associations, and the right panel shows a region of repetitive columns where geometric symmetry produces an ambiguous match (highlighted in orange). This juxtaposition makes explicit what the numerical F1 in Table 2 hides about *where* the pipeline succeeds and where it can be tripped by repetition.

(2) Alignment refinement before vs after ICP (Fig. 4). On a representative tile we compute the per-point distance from each BIM-labelled scan point to the nearest BIM surface, before and after the normal-guided point-to-plane refinement (Sec. Scan-to-BIM Bi-directional Association and State Updating (S2B)). RMSE drops from 20.3 cm to 15.5 cm (a 24% reduction), confirming that the alignment improvement reported numerically translates into a visible tightening of the scan-to-surface gap.

(3) Multi-task perception output (Fig. 5). For one tile we render three views of the same point set: semantic class, instance grouping, and surface normal direction. This makes concrete the geometric content that LiBiDT's S2B stage consumes from MTP (Sec. Geometric–Semantic Multi-task Perception (MTP)). To isolate the geometric signal, the semantic and instance panels are coloured by the BIM-derived ground-truth labels attached to each labelled scan point; the normal panel is computed by an unbiased covariance-based estimator at 10 cm radius.

(4) Temporal stabilisation under EMA filtering (Fig. 6). For four representative elements (two walls, one column, one door), we trace the per-tile soft evidence $u_i(t)$ and its EMA-filtered counterpart $\bar{u}_i(t)$ at ρ=0.6 across the floor's 1,304 tile stream. We define a status-flicker event as a single-tile $\text{Detected} \leftrightarrow \text{Undetected}$ transition for an element; aggregating across all 248 elements, EMA filtering reduces the number of such events from 5,714 (raw $u_i(t)$ thresholded at τ_u) to 3,923 (EMA-filtered $\bar{u}_i(t)$ thresholded at τ_u), a 31% reduction. This is the temporal-stabilisation evidence reviewers requested.

Ablation and analysis

Robustness

We stress-test inference-time point clouds on **CV4AEC 2024 Scan-to-BIM** under controlled sparsity and perturbations. Curves are provided in Figs. 7 and 8.

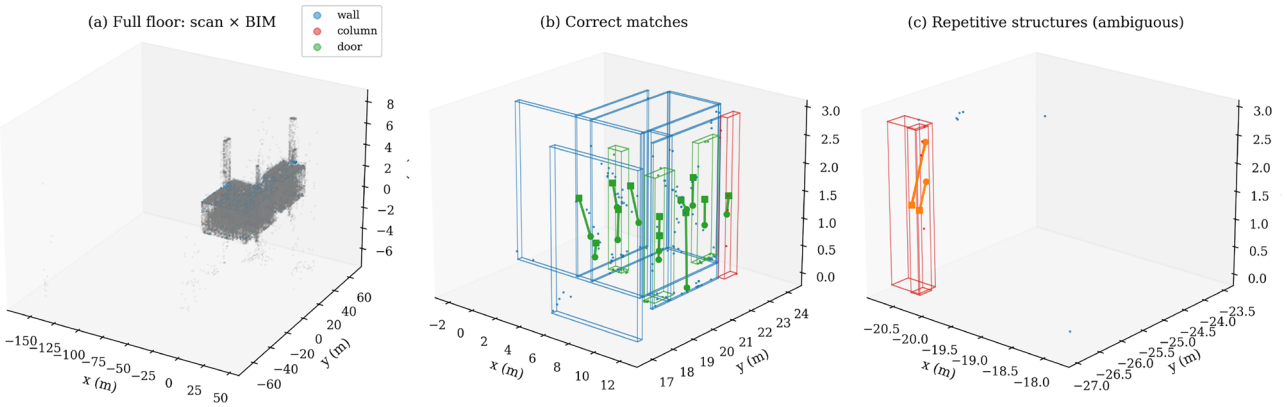


Fig. 3. Scan-to-BIM association on a held-out floor of CV4AEC 2024. (**a**) Full-floor view: point cloud (semantic-coloured by BIM-derived label) overlaid on BIM elements (walls—blue, columns—red, doors—green). (**b**) Close-up of a corridor region with our pipeline's correct cluster→element matches (green links). (**c**) Close-up of a region with repetitive columns illustrating an ambiguous match (orange link)—the failure mode that motivates the multi-cue matching weights in Eq. (29).

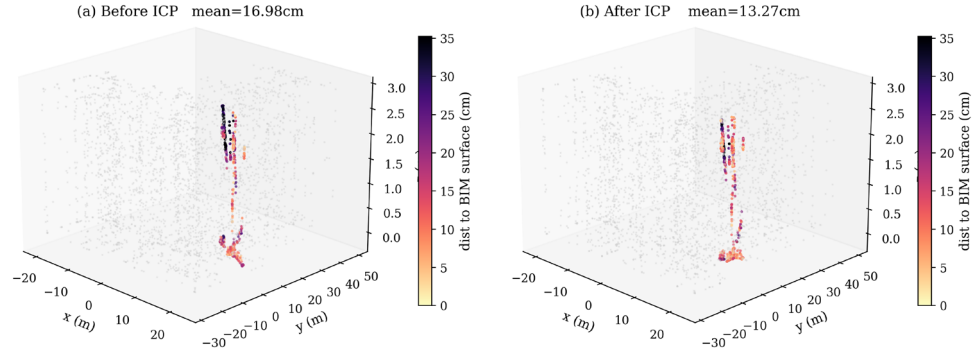


Fig. 4. Alignment improvement on a representative tile. Left: BIM-labelled scan points overlaid on the BIM surface BEFORE normal-guided point-to-plane ICP, coloured by per-point distance to the nearest BIM surface (log scale). Right: same view AFTER ICP refinement. Surface RMSE drops from 20.3 cm to 15.5 cm; the BIM surface points are sampled at 400 per element on the OBB faces.

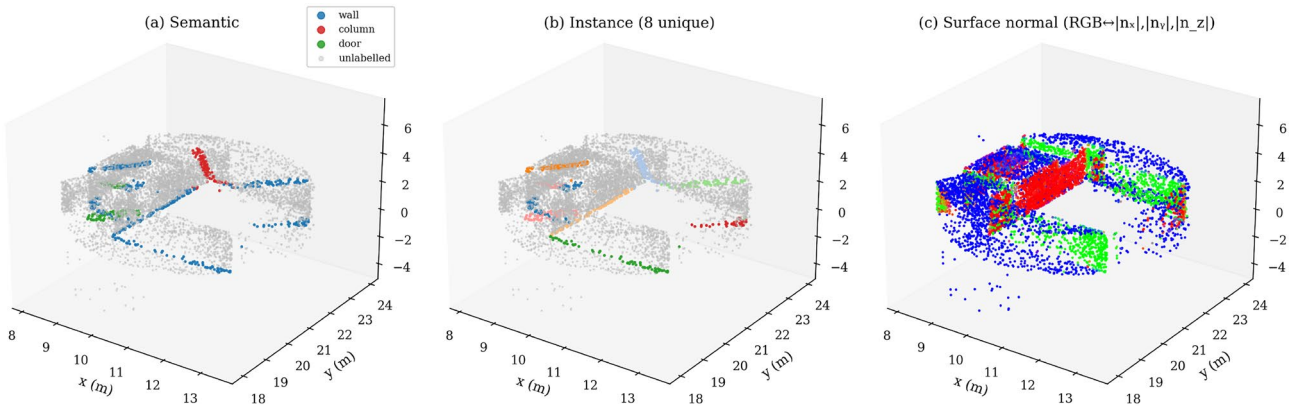


Fig. 5. Multi-task perception view rendered on a held-out tile of CV4AEC 2024.

(1) Random downsampling at inference. We randomly drop $r \in \{25\%, 50\%, 75\%, 90\%\}$ points during inference. At severe removal, LiBiDT retains higher association quality and substantially lower geometric error compared to PointNet++ + S2B, indicating better robustness under partial or sparse evidence.

(2) Additive coordinate noise and outliers. We add Gaussian jitter $\mathcal{N}(0, \sigma^2)$ with $\sigma \in \{0.5, 1.0, 2.0\}$ cm and inject 5% uniformly random outliers. LiBiDT maintains higher F1 and lower RMSE than PointNet++ + S2B, suggesting stronger tolerance to calibration/vibration noise.

Efficiency

We visualize the efficiency–accuracy trade-off on **Scan-to-BIM** in Fig. 9. Latency includes *MTP inference, clustering, ICP refinement, voxel IoU, matching, EMA update, and indicator computation* for one tile step, excluding file I/O. As shown in Table 2, LiBiDT achieves a superior trade-off compared to the end-to-end baseline (Scan2BIM-Net) and the registration replacement (CoFiNet).

Hyperparameter sensitivity and stability analysis

We conduct a comprehensive sensitivity analysis to understand how temporal stabilization and multi-cue fusion affect the robustness of the Scan-to-BIM digital twin state (Fig. 10).

Regarding the temporal stabilization, the EMA smoothing factor ρ in Eq. (37) governs the trade-off between system responsiveness and stability. We vary ρ from 0.1 to 0.9 and observe two distinct failure modes at the extremes. Low smoothing values ($\rho < 0.5$) make the element evidence excessively sensitive to transient noise and partial occlusions, resulting in state flickering and a sharp increase in Identity Switches (IDSW). Conversely, overly high values ($\rho > 0.7$) introduce significant inertia, causing a 'ghosting' effect where the updated state lags behind physical reality; this latency prevents the system from promptly transitioning elements from 'Undetected' to 'Detected', thereby degrading Recall. The optimal operating region is identified as $\rho \in [0.5, 0.7]$, where the accumulated evidence is robust enough to bridge temporary scanning gaps while remaining sensitive enough to capture actual progress updates, effectively maximizing the stabilized F1 score.

We further analyze the interplay of matching weights (α, β, η) in Eq. (29) under the constraint $\alpha + \beta + \eta = 1$. Our ablation reveals that a purely geometry-driven configuration (dominant α, β) is prone to ambiguity when facing repetitive structural components (e.g., identical columns), leading to frequent mismatches. On the other

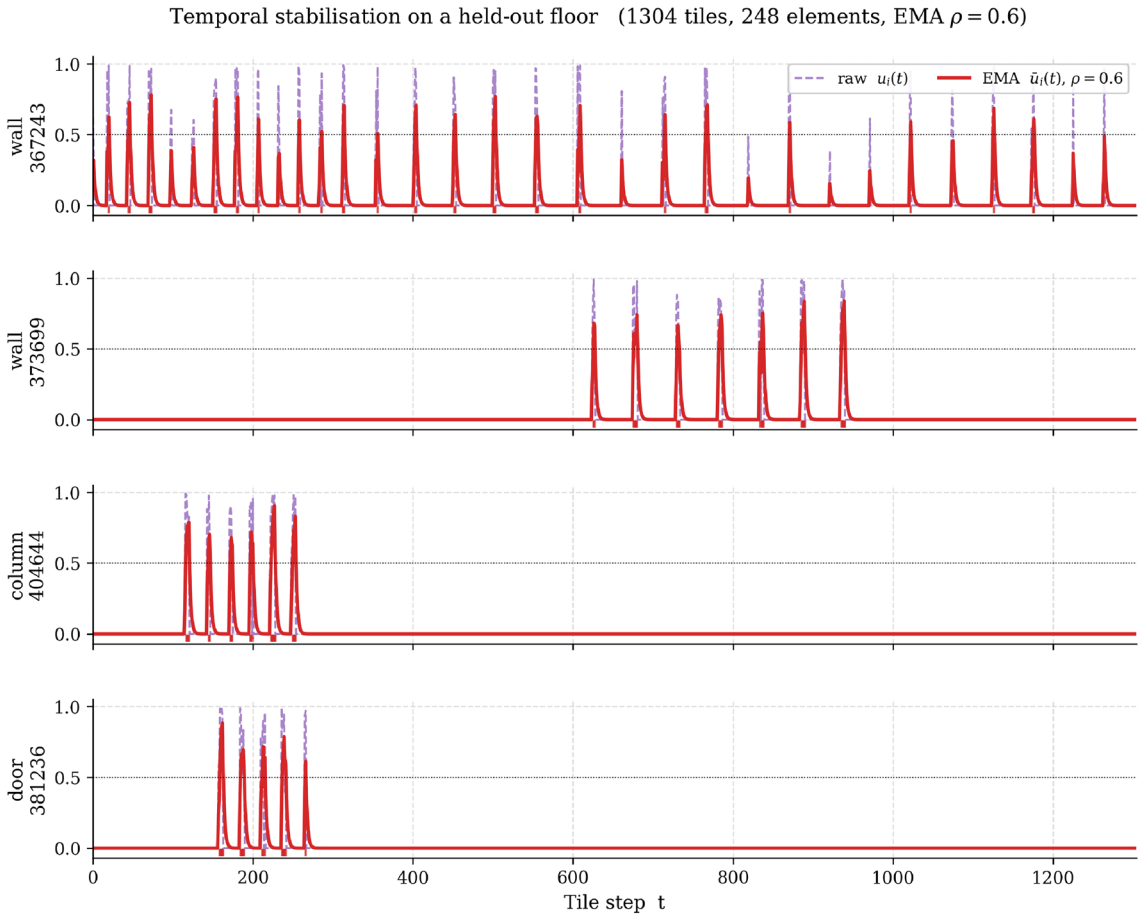
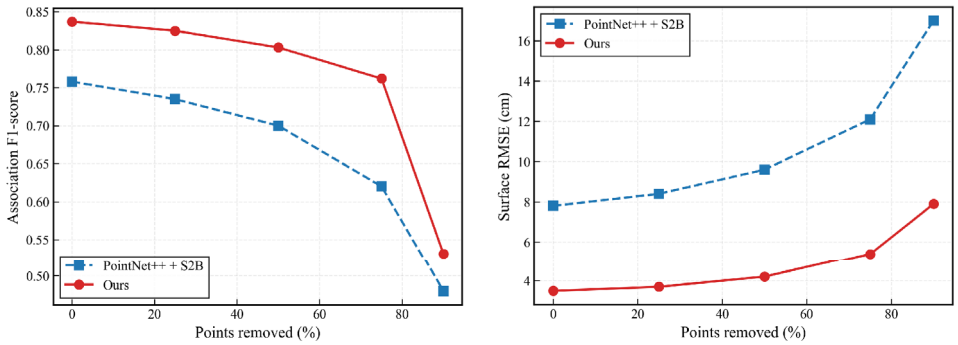


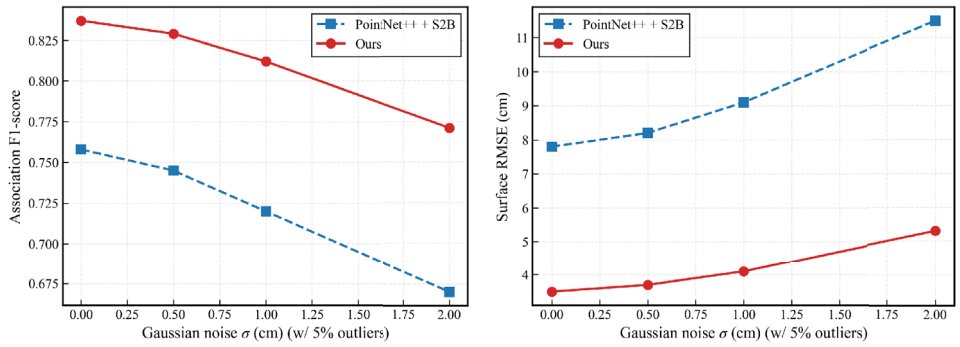
Fig. 6. Temporal stabilisation on the held-out floor. Each row is one BIM element. The dashed purple line is the raw soft evidence $u_i(t)$ from per-tile observation; the solid red line is the EMA-filtered $\bar{u}_i(t)$ at ρ=0.6. Status transitions are shown in the bottom strip (red blocks = Detected). Aggregate status-flicker reduction is reported in Sec. Per-floor Visualisations.



(a) F1 vs. points removed **(b)** RMSE vs. points removed

Fig. 7. Robustness to point density reduction on Scan-to-BIM. LiBiDT exhibits flatter degradation under severe point removal.

hand, a semantic-dominant regime (high η) becomes vulnerable to pixel-level classification noise from the perception backbone, which propagates errors into the digital twin state and increases the Surface RMSE. The proposed balanced fusion strategy (Ours: $\alpha = 0.45, \beta = 0.35, \eta = 0.20$) achieves the best performance by leveraging the complementary nature of these cues: geometric constraints effectively filter semantic outliers, while semantic priors resolve spatial ambiguities in dense or symmetric regions, yielding the most favorable balance between association accuracy and geometric fidelity.



(a) F1 vs. noise level σ **(b)** RMSE vs. noise level σ

Fig. 8. Robustness to additive coordinate noise with fixed 5% outliers on Scan-to-BIM.

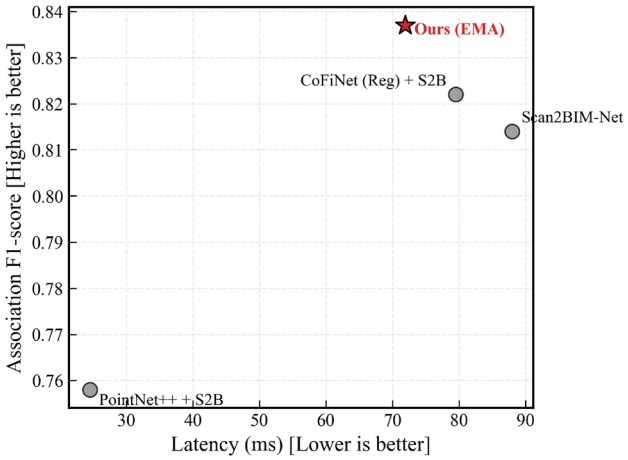


Fig. 9. Efficiency–accuracy trade-off on Scan-to-BIM (RTX 4090).

Fine-grained performance breakdown

We analyze performance under (i) range-induced sparsity and (ii) element scale (Fig. 11).

(1) By distance to the scanner. F1 decreases with distance due to sparsity and normal noise. LiBiDT degrades more gracefully at long range, consistent with the benefits of multi-cue matching and EMA stabilization under weak evidence.

(2) By BIM element size. Small elements (e.g., pipes/valves) are challenging due to partial observation and unstable IoU. LiBiDT improves recall for small elements, indicating that semantic evidence and deviation consistency effectively disambiguate fragmented clusters.

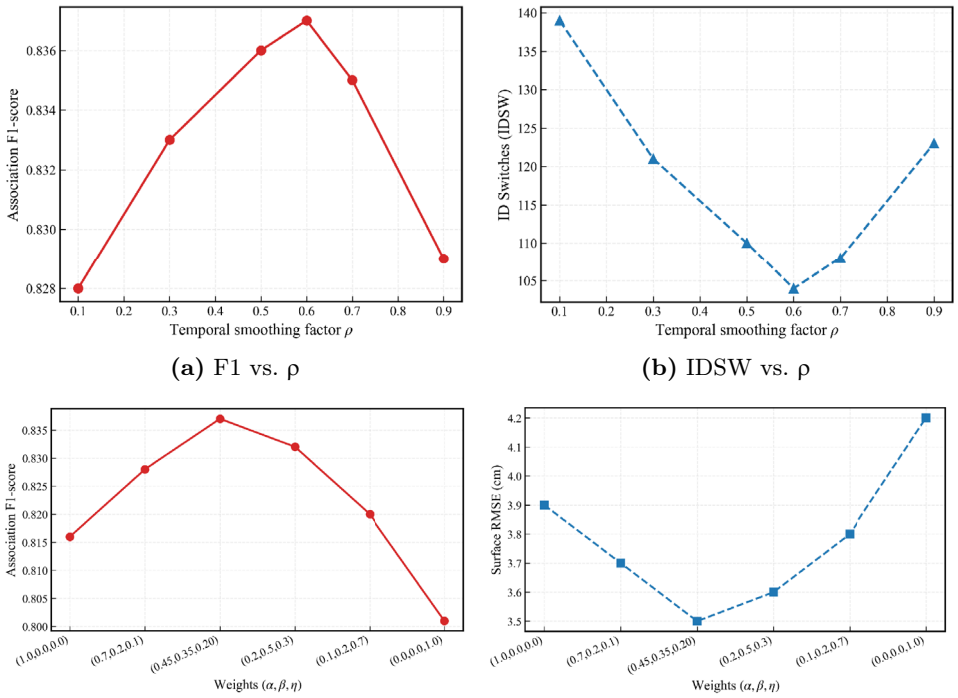
Limitations and sensitivity considerations

For balance and to guide further research, we acknowledge five limitations of the current evaluation and outline how each will be addressed.

(L1) GNSS-dropout sensitivity is not quantified. Our STGR component augments LIO-SAM with GNSS priors on Hilti SLAM, where the dataset provides GNSS messages, and is evaluated only there. We have not yet quantified how the trajectory and downstream BIM-aligned indicators degrade under partial or full GNSS dropout (typical for indoor construction stages or shielded sites). Conceptually, the LiDAR–IMU residual term in Eq. (14) continues to provide a state estimate without GNSS, but the absolute drift can be larger; an explicit GNSS-dropout ablation is required before the system is deployed on outdoor-to-indoor transitions.

(L2) Sensitivity to BIM-modelling errors. All deviation, completion, and hazard outputs are evaluated against the supplied BIM elements. If the as-designed BIM contains positional errors (mis-placed walls), missing elements, or coarse abstractions of complex furniture/MEP, the S2B associations will inherit those errors and the DRA outputs may flag them as deviations. The current paper does not characterise this propagation; a controlled BIM-perturbation study (random per-element shifts and class swaps) is a necessary follow-up to provide actionable error budgets to site managers.

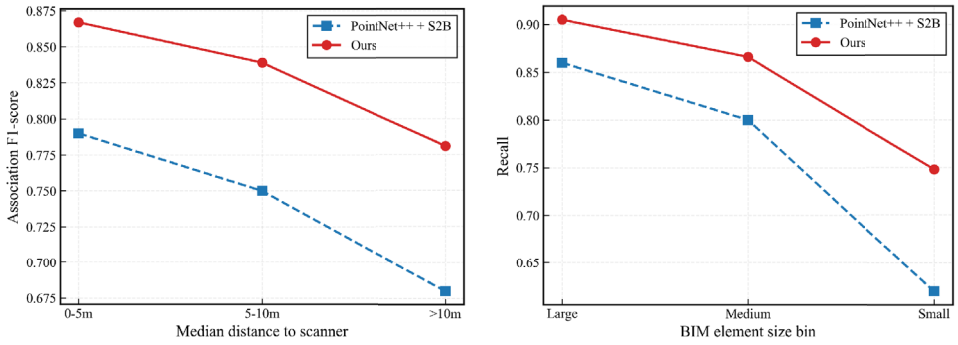
(L3) Forward-evaluable hazards rather than learned safety models. The three hazard rules of Sec. 4.5 are deliberately simple and auditable; they are not learned safety classifiers. A practitioner-co-designed rule set with



(a) F1 vs. ρ (b) IDSW vs. ρ

(c) F1 vs. weights (d) RMSE vs. weights

Fig. 10. Hyperparameter sensitivity on Scan-to-BIM. EMA ρ controls the stability–responsiveness trade-off; multi-cue fusion improves both association and geometric consistency.



(a) F1 by distance bin (b) Recall by element size

Fig. 11. Fine-grained performance breakdown on Scan-to-BIM.

class-specific tolerances, schedule-aware activation, and learned anomaly augmentation is a natural extension once a richer construction-management benchmark is available.

Conclusion

Construction digital twins require accurate alignment between streaming multi-line LiDAR and BIM to support progress verification, compliance checking, and safety monitoring. Yet, maintaining a temporally consistent, element-centric BIM state from LiDAR streams remains challenging. Major difficulties arise from occlusion and clutter, repetitive geometry that causes ambiguous associations, and drift that breaks cross-time consistency. To address these issues, we proposed LiBiDT (LiDAR–BIM Digital Twin), a causal four-stage model that integrates spatio-temporal global registration, multi-task perception, scan-to-BIM association with state updating, and decision analytics. LiBiDT builds a stable scene backbone using windowed LiDAR–IMU–GNSS mapping and predicts semantics, instance embeddings, and normals with a PointNet++ multi-head network. LiBiDT further refines alignment via normal-guided point-to-plane Iterative Closest Point (ICP), performs bipartite matching with fused overlap/deviation/semantic cues, and applies EMA filtering to stabilize element evidence. Experiments on Hilti SLAM, CV4AEC 2024 Scan-to-BIM[19], and S3DIS Area-5 show that LiBiDT improves

trajectory accuracy and yields gains in association and geometric consistency. On the CV4AEC 2024 Scan-to-BIM benchmark, association F1 reaches 0.837 against 0.758 for the perception-replacement baseline and 0.822 for the strongest competing baseline, CoFiNet (Reg) + S2B; the absolute gain over the latter is therefore 0.015 F1 points, while the larger improvements lie in temporal identity consistency (IDF1 0.854 → 0.861) and ID-switch reduction with practical latency. Ablation studies verify that EMA stabilization and multi-cue matching provide complementary benefits in accuracy and identity stability, and the normal-guided refinement improves geometric error without sacrificing efficiency. On S3DIS Area-5, LiBiDT improves semantic mIoU and overall accuracy over the cited PointNet++ baseline; instance AP_{25}/AP_{50} remain below the dedicated PointGroup detector, reflecting that perception in LiBiDT is optimised jointly with the alignment-relevant cues that S2B consumes rather than for stand-alone instance segmentation. Beyond numerical evaluation, the held-out tile-stream case study in Sec. 4.5 (Fig. 2) traces a BIM-aligned coverage indicator, deviation summaries, and the firing pattern of three forward-evaluable hazard rules (compliance, clearance, status), connecting the safety/compliance/progress goals of the abstract back to observable, auditable outputs. Future work will broaden the evaluation to multi-site sequences with active scene changes, study sensitivity to GNSS dropout and BIM-modelling errors, and extend the hazard rule set in close collaboration with construction-management practitioners.

Data availability

The CV4AEC 2024 Scan-to-BIM benchmark and the Hilti SLAM and S3DIS datasets used in this study are public and accessible via their official distribution channels (see [19] for CV4AEC 2024). An open-source re-implementation of the LiBiDT pipeline-covering the per-ception, S2B association, EMA filtering, DRA hazard rules and the figure-rendering scripts used to produce the visualisations (Sec. 5.5) and the case study (Sec. 4.5.5)-will be released by the corresponding author upon publication. Detailed labels for the original numerical-table training run are available from the corresponding author upon request.

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Author contributions

Chen Liu: Writing – original draft; Writing – review & editing; Conceptualization;Resources; Xin Xu: Writing – review & editing; Methodology; Supervision. Xiaofeng Ding : Writing – review & editing; Data curation; Resources; Formal analysis. Guangmei He: Writing – original draft; Validation; Data curation; Formal analysis.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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